

PAC-Bayesian Analysis of the Weighted Majority Vote

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Weighted Majority Vote

- Fundamental technique for combining predictions of multiple classifiers
- Used in Random Forests, Boosting, and other techniques
- Can be applied with heterogeneous classifiers
- Wins most ML competitions
- Can be used for derandomization of PAC-Bayes

Key power - Cancellation of errors effect

- *If* the errors are independent, they average out

Main theoretical questions

- Generalization bounds
- Optimization of weights

Weighted Majority Vote – Formal Definition

- Let ρ be a distribution on $\mathcal{H}$
- In the binary case ($\mathcal{Y} = \{\pm 1\}$)
 - $MV_{\rho}(X) = \text{sign}(\sum_h h(X)\rho(h))$
 - The label getting the higher weight
 - Ties resolved arbitrarily
- In the multiclass case (finite $\mathcal{Y}$)
 - $MV_{\rho}(X) = \arg \max_y \sum_{h:h(X)=y} \rho(h) = \arg \max_y \mathbb{E}_{h \sim \rho}[\mathbb{I}(h(X) = y)]$
 - The label getting the maximal weight
 - Ties resolved arbitrarily

Majority Vote – some intuition – binary prediction

- Example: Anticorrelated errors, $L(h_i) < \frac{1}{2}$

	$\mathcal{X}$		
h_1	correct	correct	incorrect
h_2	correct	incorrect	correct
h_3	incorrect	correct	correct
MV	correct	correct	correct

- Example: Anticorrelated errors, $L(h_i) > \frac{1}{2}$

	$\mathcal{X}$		
h_1	correct	incorrect	incorrect
h_2	incorrect	correct	incorrect
h_3	incorrect	incorrect	correct
MV	incorrect	incorrect	incorrect

- Example: Correlated errors

	$\mathcal{X}$		
h_1	correct	incorrect	incorrect
h_2	correct	incorrect	incorrect
h_3	correct	incorrect	incorrect
MV	correct	incorrect	incorrect

Majority Vote – some intuition - continued

- If majority vote erred, then at least a ρ -weighted half of the classifiers erred
- Conclusion #1: the expected error of the weighted majority vote is at most twice the ρ -weighted average of the loss of individual classifiers
$$L(\text{MV}_\rho) \leq 2\mathbb{E}_{h \sim \rho}[L(h)]$$
- Conclusion #2: *If* the errors are independent and error rates are identical and smaller than 0.5, so that $L(h) = 0.5 - \varepsilon$, then
 - $L(\text{MV}_\rho) \leq \mathbb{P}(\text{"more than } \rho\text{-weighted half errs"}) \leq e^{-2M\varepsilon^2}$
 - Converges to zero as $|\mathcal{H}| = M$ grows
 - “The more the better” under the assumption of being better than 0.5 chance
 - Unlike in selection!

Construction of ensembles for weighted majority vote

- Random forests
 - Use bagging (training on random subsamples) to increase independence of errors
- Boosting
 - Increase anticorrelation of errors through focus on residual errors

Majority vote – mid-summary

- If $L(h) < 0.5$, then including h in the majority vote helps
- Most helpful when the errors are anticorrelated, helpful when the errors are independent, not helpful if the errors are correlated
- If $L(h) < 0.5$ for all h and independent or anticorrelated errors, then $L(MV_\rho)$ converges to zero as the number of classifiers grows
- The more the better (unlike in selection!)
 - But remember that $L(h)$ is unknown, and in estimation of $L(h)$ the issue of selection comes back
- Never worse than twice the average error of individual classifiers
- Techniques for constructing good ensembles:
 - Bagging
 - Train on random subsamples to increase independence of errors
 - Boosting
 - Train on residual errors to increase anticorrelation of errors
- Challenges:
 - True error rates are unknown
 - True correlation is unknown
 - Weight tuning

Bounding $L(MV_\rho)$

- Basic observation:
 - If majority vote erred, then at least a ρ -weighted half of the classifiers erred
 - $\ell(MV_\rho(X), Y) \leq \mathbb{I} \left(\underbrace{\mathbb{E}_{h \sim \rho} [\mathbb{I}(h(X) \neq Y)]}_{\rho\text{-weighted mass of errors}} \geq 0.5 \right)$

- Basic bound:
 - $\underbrace{L(MV_\rho)}_{\substack{\text{expected loss of} \\ \rho\text{-weighted} \\ \text{majority vote}}} = \mathbb{E}_{(X,Y) \sim D} [\ell(MV_\rho(X), Y)]$

$$\leq \mathbb{E}_{(X,Y) \sim D} \left[\mathbb{I} \left(\underbrace{\mathbb{E}_{h \sim \rho} [\mathbb{I}(h(X) \neq Y)]}_{\substack{\rho\text{-weighted mass} \\ \text{of errors on } (X,Y)}} \geq 0.5 \right) \right]$$

$$= \mathbb{P}_{(X,Y) \sim D} \left(\underbrace{\mathbb{E}_{h \sim \rho} [\mathbb{I}(h(X) \neq Y)]}_{\substack{\rho\text{-weighted mass} \\ \text{of errors on } (X,Y)}} \geq 0.5 \right)$$

First order oracle bound (“folk theorem”)

- Theorem: First order oracle bound

$$L(\text{MV}_\rho) \leq 2 \underbrace{\mathbb{E}_{h \sim \rho}[L(h)]}_{\substack{\text{expected loss of} \\ \rho\text{-weighted} \\ \text{randomized classifier}}}$$

- Proof:

- The basic bound: $L(\text{MV}_\rho) \leq \mathbb{P}_{(X,Y) \sim D}(\mathbb{E}_{h \sim \rho}[\mathbb{I}(h(X) \neq Y)] \geq 0.5)$

- Take $Z = \underbrace{\mathbb{E}_{h \sim \rho}[\mathbb{I}(h(X) \neq Y)]}_{\rho\text{-weighted mass of errors on } (X,Y)}$

$$\begin{aligned} L(\text{MV}_\rho) \leq \mathbb{P}(Z \geq 0.5) &\stackrel{\text{Markov}}{\leq} 2\mathbb{E}_{(X,Y) \sim D}[Z] \\ &= 2\mathbb{E}_{(X,Y) \sim D} \left[\mathbb{E}_{h \sim \rho}[\mathbb{I}(h(X) \neq Y)] \right] = 2\mathbb{E}_{h \sim \rho} \left[\mathbb{E}_{(X,Y) \sim D}[\mathbb{I}(h(X) \neq Y)] \right] = 2\mathbb{E}_{h \sim \rho}[L(h)] \end{aligned}$$

First order empirical bound

$$\underbrace{L(\text{MV}_\rho) \leq 2\mathbb{E}_{h \sim \rho}[L(h)]}_{\text{First order oracle bound}} \leq 2 \left(\underbrace{\frac{\mathbb{E}_{h \sim \rho}[\hat{L}(h, S)]}{1 - \frac{\lambda}{2}} + \frac{\text{KL}(\rho \parallel \pi) + \ln \frac{2\sqrt{n}}{\delta}}{n\lambda \left(1 - \frac{\lambda}{2}\right)}}_{\text{PAC-Bayesian bound on } \mathbb{E}_{h \sim \rho}[L(h)]} \right)$$

- Advantages:
 - Reasonably tight
- Disadvantages:
 - The oracle bound ignores correlations of errors (the main power of MV)
 - Optimization of the empirical bound leads to deterioration of the test error

Second order oracle bound

- Define *tandem loss* for pairs of classifiers h, h'

- $\ell(h(X), h'(X), Y) = \underbrace{\mathbb{I}(h(X) \neq Y) * \mathbb{I}(h'(X) \neq Y)}_{\text{tandem loss}}$

- Counts an error if both h and h' err on (X, Y)

- Expected tandem loss:

- $L(h, h') = \mathbb{E}_{(X,Y) \sim D} [\ell(h(X), h'(X), Y)]$

- Empirical tandem loss:

- $\hat{L}(h, h', S) = \frac{1}{n} \sum_{i=1}^n \ell(h(X_i), h'(X_i), Y_i)$

- Theorem: Second order oracle bound

$$L(MV_\rho) \leq 4 \mathbb{E}_{(h,h') \sim \rho^2} \left[\underbrace{L(h, h')}_{\substack{\text{expected} \\ \text{tandem loss}}} \right]$$

- Proof:

- Second order Markov's inequality:

$$\mathbb{P}(Z \geq \varepsilon) \leq \mathbb{P}(Z^2 \leq \varepsilon^2) \leq \mathbb{E}[Z^2] / \varepsilon^2$$

- As before, take $Z = \mathbb{E}_{h \sim \rho} [\mathbb{I}(h(X) \neq Y)]$

- $\underbrace{L(MV_\rho)}_{\text{The basic bound}} \leq \mathbb{P}(Z \geq 0.5)$

$$\leq 4 \mathbb{E}_{(X,Y) \sim D} [Z^2]$$

$$= 4 \mathbb{E}_{(X,Y) \sim D} [(\mathbb{E}_{h \sim \rho} [\mathbb{I}(h(X) \neq Y)])^2]$$

$$= 4 \mathbb{E}_{(X,Y) \sim D} [\mathbb{E}_{(h,h') \sim \rho^2} [\underbrace{\mathbb{I}(h(X) \neq Y) * \mathbb{I}(h'(X) \neq Y)}_{\text{tandem loss}}]]$$

$$= 4 \mathbb{E}_{(h,h') \sim \rho^2} [\mathbb{E}_{(X,Y) \sim D} [\underbrace{\mathbb{I}(h(X) \neq Y) * \mathbb{I}(h'(X) \neq Y)}_{\text{tandem loss}}]]$$

$$= 4 \mathbb{E}_{(h,h') \sim \rho^2} [L(h, h')]$$

Second order empirical bound

$$\begin{aligned} L(MV_\rho) &\leq 4\mathbb{E}_{(h,h')\sim\rho^2}[L(h,h')] \\ &\leq 4\left(\underbrace{\frac{\mathbb{E}_{(h,h')\sim\rho^2}[\hat{L}(h,h',S)]}{1-\frac{\lambda}{2}} + \frac{\text{KL}(\rho^2\|\pi^2)+\ln\frac{2\sqrt{n}}{\delta}}{n\lambda\left(1-\frac{\lambda}{2}\right)}}_{\text{PAC-Bayesian bound on } \mathbb{E}_{(h,h')\sim\rho^2}[L(h,h')]} \right) \\ &= 4\left(\frac{\mathbb{E}_{(h,h')\sim\rho^2}[\hat{L}(h,h',S)]}{1-\frac{\lambda}{2}} + \frac{2\text{KL}(\rho\|\pi)+\ln\frac{2\sqrt{n}}{\delta}}{n\lambda\left(1-\frac{\lambda}{2}\right)}\right) \end{aligned}$$

- Advantages:
 - Takes correlation of errors into account
 - Minimization does not deteriorate the test error
- Disadvantages:
 - The tandem loss is harder to estimate (both computationally and statistically), therefore the bound is often weaker than the first order bound
 - But it is still a better bound to optimize, because it does not deteriorate the test error

Comparison of the oracle bounds

- First order oracle bound

$$L(MV_\rho) \leq 2\mathbb{E}_{h \sim \rho}[L(h)]$$

- Second order oracle bound

$$L(MV_\rho) \leq 4\mathbb{E}_{(h,h') \sim \rho^2}[L(h, h')]$$

- Comparison

- Assume equal $L(h)$ for all h
- $|\mathcal{H}| = M$
- ρ - uniform

- Worst case: identical error regions

- $L(MV_\rho) = L(h)$
- $FO(\rho) = 2L(h)$
- $SO(\rho) = 4L(h)$

- Best case: disjoint error regions, $M \geq 3$, $L(h) < \frac{1}{2}$

- $L(MV_\rho) = 0$
- $FO(\rho) = 2L(h)$
- $SO(\rho) = 4L(h)/M$

- Independent case: independent errors, $L(h) < \frac{1}{2}$

- $L(MV_\rho) \xrightarrow{M \rightarrow \infty} 0$
- $FO(\rho) = 2L(h)$
- $SO(\rho) = 4 \left(L(h)^2 + \frac{1}{M} L(h)(1 - L(h)) \right)$
 - For large M approximately $4L(h)^2$
 - For $L(h) < 0.5$ tighter than $FO(\rho)$

Ensemble construction

- Generate M random splits of the data $S = T_h \cup S_h$, such that $T_h \cap S_h = \emptyset$
 - A natural construction in random forests
- Train h on T_h and compute $\hat{L}(h, S_h)$
 - Unbiased estimate
- For tandem loss compute $\hat{L}(h, h', S_h \cap S_{h'})$
 - Unbiased estimate

Modified empirical bounds

- $n = |S_h|$
- $m = \min_{h,h'} |S_h \cap S_{h'}|$
- $\pi(h) = \frac{1}{M}$

- Modified empirical bounds

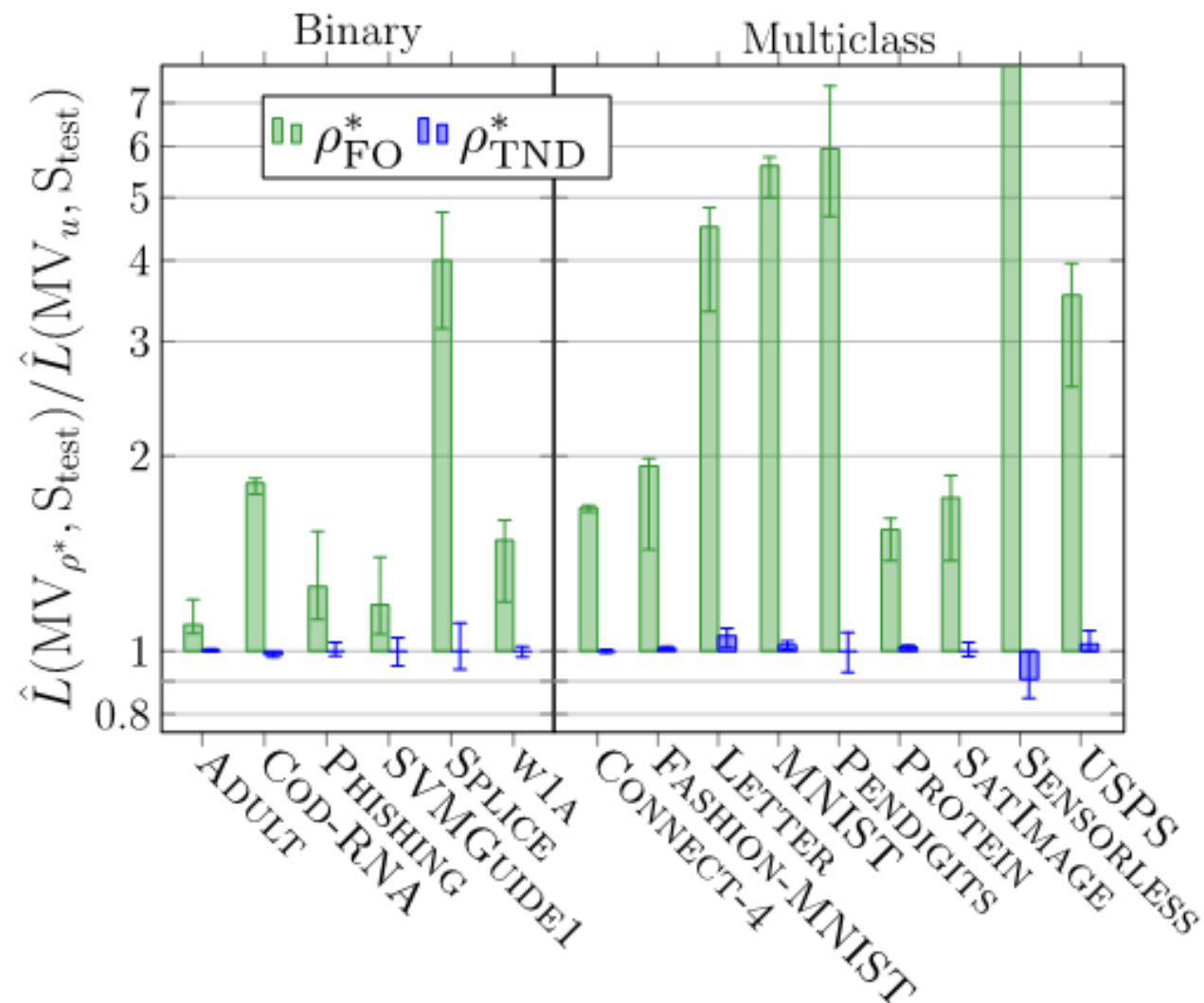
$$\mathbb{E}_{h \sim \rho}[L(h)] \leq \frac{\mathbb{E}_{h \sim \rho}[\hat{L}(h, S_h)]}{1 - \frac{\lambda}{2}} + \frac{\text{KL}(\rho \parallel \pi) + \ln \frac{2\sqrt{n}}{\delta}}{n\lambda \left(1 - \frac{\lambda}{2}\right)}$$

$$\mathbb{E}_{(h,h') \sim \rho^2}[L(h, h')] \leq \frac{\mathbb{E}_{(h,h') \sim \rho^2}[\hat{L}(h, h', S \cap S_{h'})]}{1 - \frac{\lambda}{2}} + \frac{2\text{KL}(\rho \parallel \pi) + \ln \frac{2\sqrt{m}}{\delta}}{m\lambda \left(1 - \frac{\lambda}{2}\right)}$$

- Advantages of empirical SO vs. FO:
 - $\hat{L}(h, h', S \cap S_{h'}) \leq \hat{L}(h, S_h)$
 - Boosted by fast convergence rates
- Disadvantages of empirical SO vs. FO:
 - $m \leq n$
 - $2\text{KL}(\rho \parallel \pi)$
- Relative power depends on the data

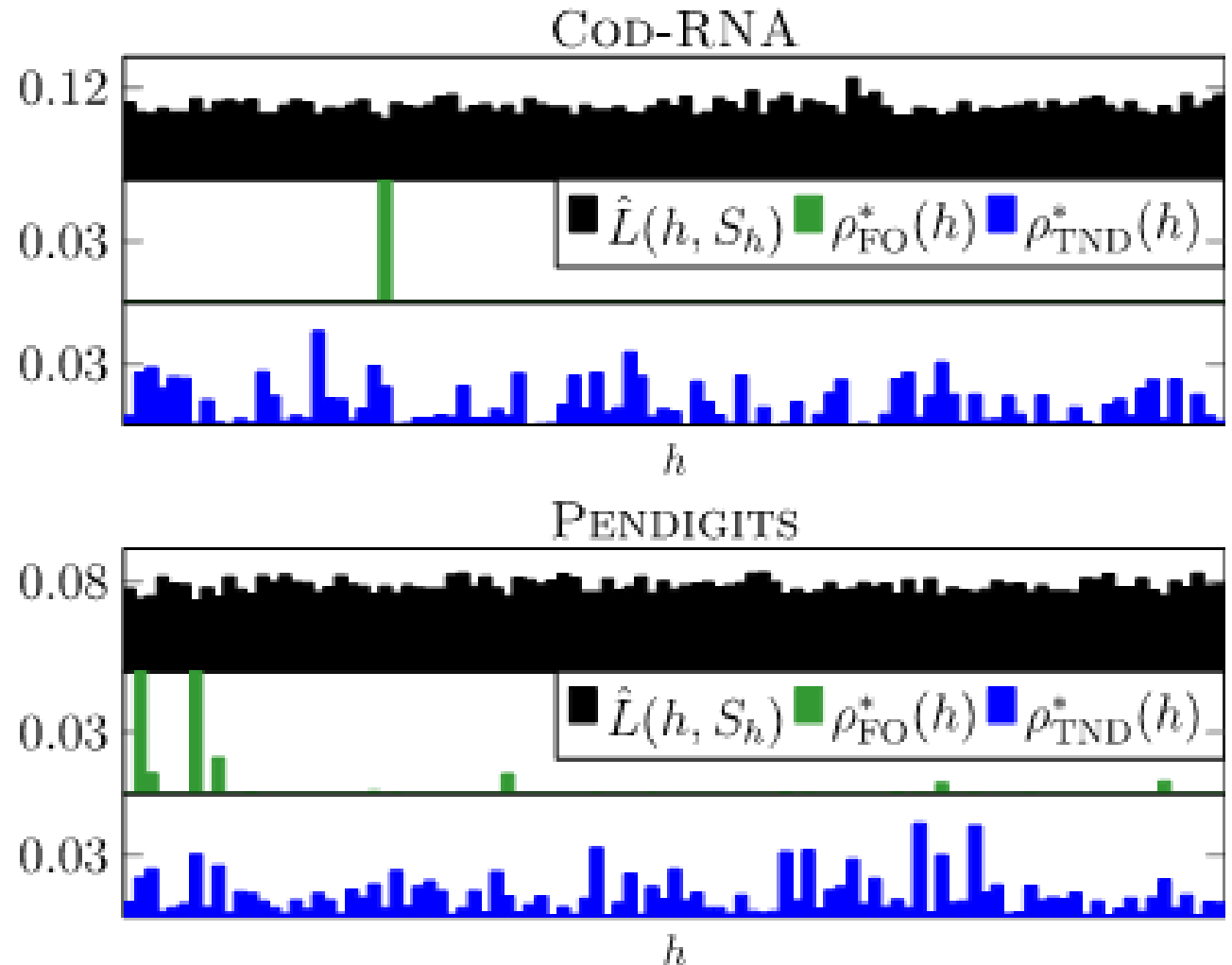
Empirical evaluation

Test error of optimized majority vote over uniformly weighted majority vote baseline for the **first order [FO]** and the **second order [TND]** bound (the lower the better)



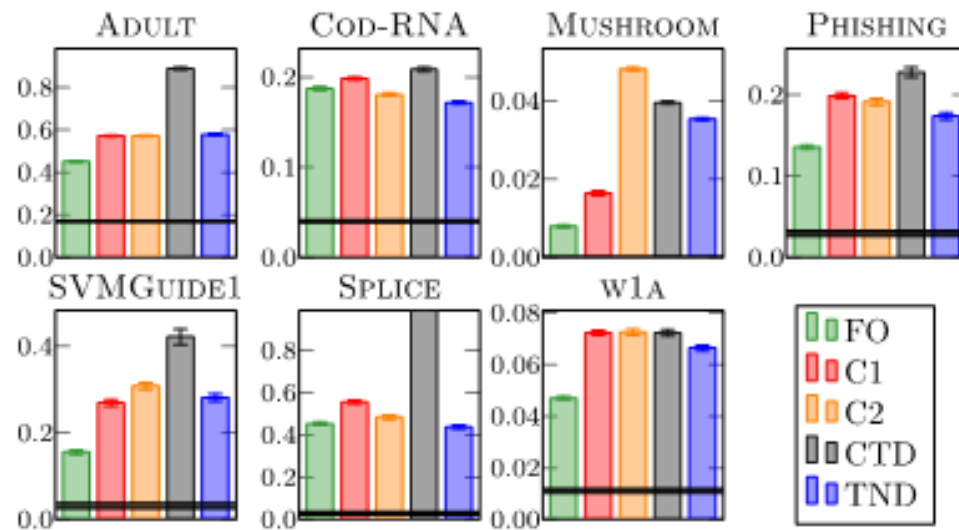
Empirical evaluation

The optimized weights ρ^* generated by the **first order [FO]** and the **second order [TND]** bound.

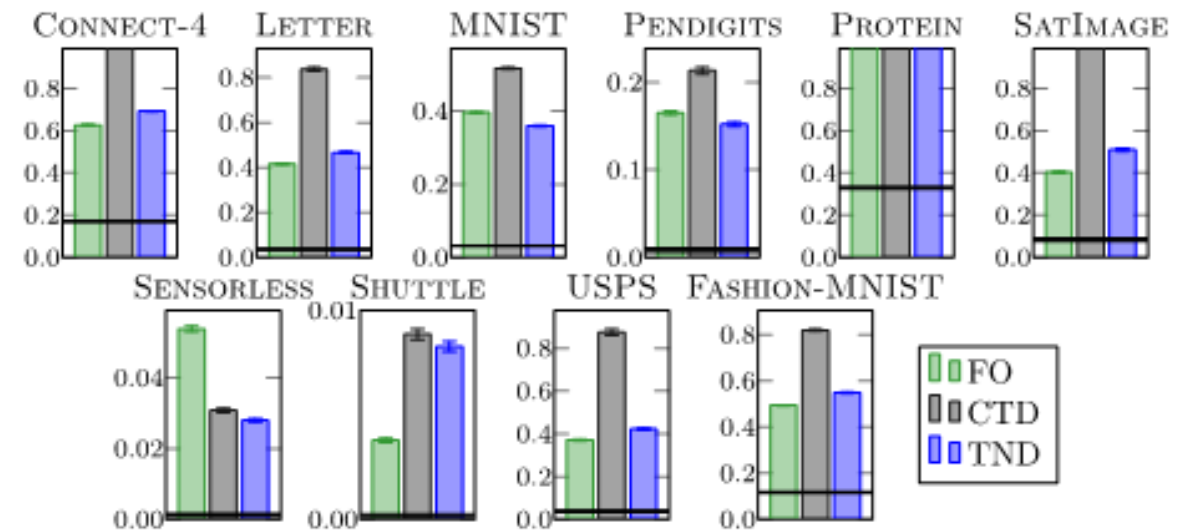


Empirical evaluation – bounds for a uniformly weighted ensemble (not optimized ρ)

Binary classification



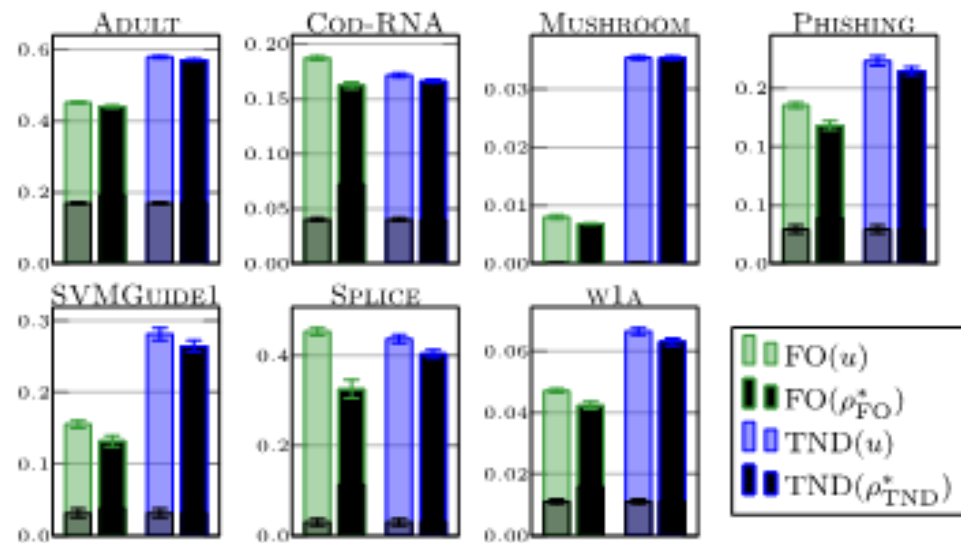
Multiclass classification



- FO – First Order
- TND – the new second order bound
- C1, C2, CTD – alternative second order bounds (Germain et al., 2015) – cannot be optimized!
- More details in Masegosa et al., 2020.

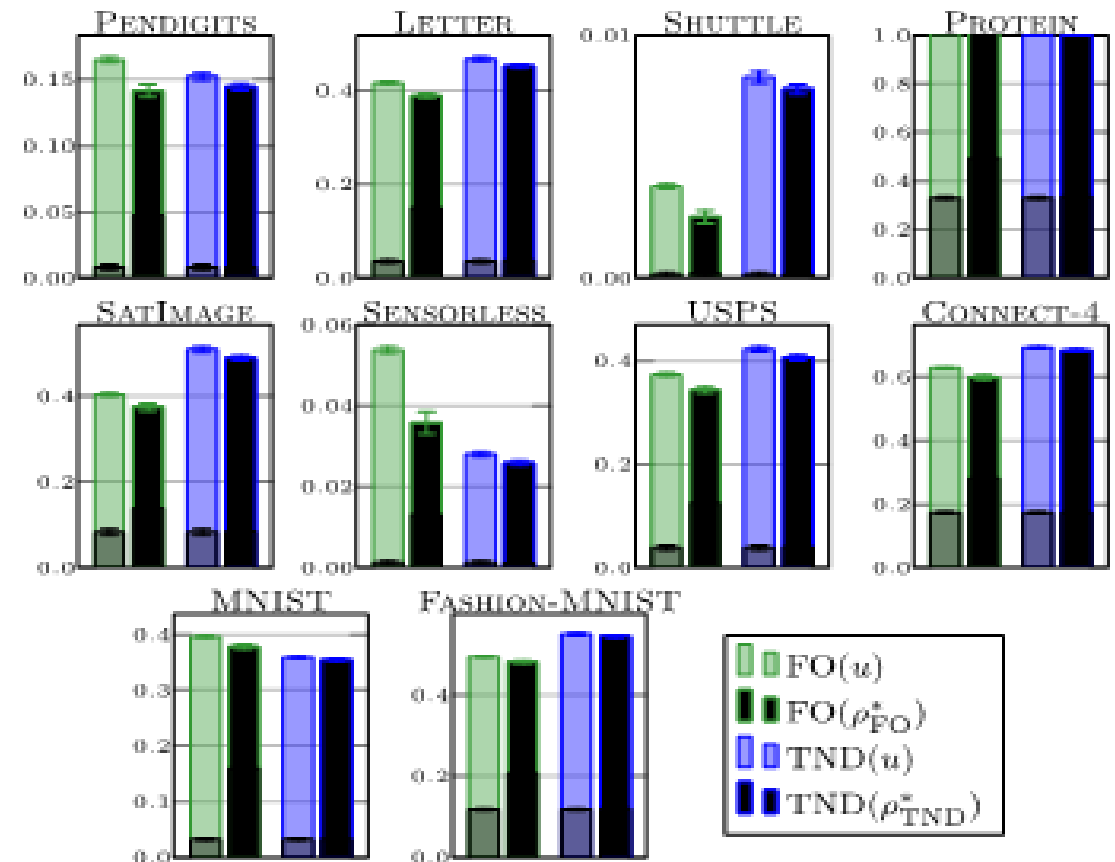
Empirical evaluation – bounds before and after optimization

Binary classification



- FO – First Order
- TND – second order bound

Multiclass classification



Summary

- The basic bound:

- $L(MV_\rho) \leq \mathbb{P}_{(X,Y) \sim D}(\mathbb{E}_{h \sim \rho}[\mathbb{I}(h(X) \neq Y)] \geq 0.5)$

- First order bound

- $$\underbrace{L(MV_\rho) \leq 2\mathbb{E}_{h \sim \rho}[L(h)]}_{\text{First order oracle bound}} \leq 2 \left(\underbrace{\frac{\mathbb{E}_{h \sim \rho}[\hat{L}(h,S)]}{1 - \frac{\lambda}{2}} + \frac{\text{KL}(\rho \parallel \pi) + \ln \frac{2\sqrt{n}}{\delta}}{n\lambda(1 - \frac{\lambda}{2})}}_{\text{PAC-Bayesian bound on } \mathbb{E}_{h \sim \rho}[L(h)]} \right)$$

- Ignores correlations of errors
 - Minimization degrades the test error

- Second order bound

- $$\underbrace{L(MV_\rho) \leq 4\mathbb{E}_{(h,h') \sim \rho^2}[L(h,h')]}_{\text{Second order oracle bound}} \leq 4 \left(\frac{\mathbb{E}_{(h,h') \sim \rho^2}[\hat{L}(h,h',S)]}{1 - \frac{\lambda}{2}} + \frac{2\text{KL}(\rho \parallel \pi) + \ln \frac{2\sqrt{n}}{\delta}}{n\lambda(1 - \frac{\lambda}{2})} \right)$$

- Takes correlations of errors into account
 - Does not deteriorate the test error
 - May be weaker than the first order bound

- Weighted majority vote is helpful whenever the expected errors of individual classifiers are below 0.5 and the errors are uncorrelated or weakly correlated

- “The more the better”, unlike in selection (but selection affects estimation of $L(h)$)

- Techniques for constructing ensembles

- Bagging
 - Boosting

- Follow-ups:

- Bounds based on Chebyshev-Cantelli instead of second order Markov
 - Wu et al., NeurIPS, 2021
 - Chebyshev-Cantelli + PAC-Bayes-split-kl
 - Wu and Seldin, NeurIPS, 2022

Open question

- We lose a factor of 2 or 4 when we go via the oracle bounds
 - $L(\text{MV}_\rho) \leq 2\mathbb{E}_{h \sim \rho}[L(h)]$
 - $L(\text{MV}_\rho) \leq 4\mathbb{E}_{(h, h') \sim \rho^2}[L(h, h')]$
- And then we bound the risk of a randomized classifier in terms of its empirical risk, but a randomized classifier is a priori not as good as a majority vote
- Open question:
 - Could we do better by directly bounding the distance between $L(\text{MV}_\rho)$ and (some form of) its empirical counterpart that would hold for all ρ ?
 - And provide a computationally efficient optimization with respect to ρ ?